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Linguistic Essentials

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Parts of Speech and Morphology

Linguists group words of a language into *syntactic* or *grammatical* categories, also known as *parts of speech*.

Open or *lexical* categories are most common, have many members

- Example: Children eat sweet candy.
- *Nouns*: refer to things (children, candy)
- *Verb*: expresses action (eat)
- *Adjective*: describes properties of nouns (sweet)

Closed or *functional* categories only have a few members

- Example: prepositions and determiners
- Contains words such as *of*, *on*, *the*, *a*

Parts of Speech and Morphology

Various parts of speech for a word are listed in a *dictionary* or *lexicon*.

Often we want to *tag* words of a sentence's with their POS (parts of speech).

Word categories are often related by *morphological processes*

- For example, adding an *s* turns *dog* (singular) into *dogs* (plural)
- Many unknown words are morphologically related to known words
- Modelling morphology is especially important in highly inflected languages like Finnish

Major types of morphological processes:

- Inflection: systematic modifications of a root form using prefixes and suffixes; does not change word class or meaning significantly, but may depend on language/context

Morphological Processes

Inflection: systematic modifications of a *root form* using *prefixes* and *suffixes*

- Does not change word class or meaning significantly
  - All inflectional forms of word are grouped as a single *lexeme*

Derivation: more change of syntactical category, often changes meaning

- Derivation of *wide* into *widely* changes adjective into adverb
- Less systematic: for example, *difficultly* is not an English word
- Other derivations: *-en* suffix transforms adjectives to verbs, e.g., bright → brighten
- *-able* suffix turns verbs into adjectives, e.g., market → marketable
- *-er* suffix turns verbs into nouns, e.g., dig to digger (as always there are exceptions and ….)

Compounding: combine two or more words into a new word

- Example: *downtown, overtake*
- Some compounds are written as two words, but mean one thing: *tea kettle, disk drive*

Nouns

- Nouns refer to entities in the world.
  - Examples: dog, tree, house
- Nouns can usually be pluralized by adding an *s* suffix
  - Words that don't follow this pattern, like *women*, are *irregular*
- Some languages have *case inflections*
  - In some languages, nouns are inflected depending on whether they are subjects or objects.
  - English has cases, but not many case inflections
  - *Genitive* case: the *'s* in *student's* indicates possession
  - *'s* is a *clitic* or *phrasal affix*, not an inflection

Pronouns

Pronouns: a small class of words which refer to an object based on context.

- Pronouns are inflected based on whether they are subjects or objects.
  - Nominative or subject case: I, he, she
  - Accusative or object case: me, him, her
- Possessive pronouns:
  - "A friend of *mine*"
- Reflexive pronouns refers to self:
  - *Mary saw herself in the mirror. Herself* is reflexive, refers to Mary.
  - *Mary saw her in the mirror. Her* is not reflexive, cannot refer to Mary.
  - Reflexive pronouns are often referred to as *anaphors*.

Determiners and Adjectives

- Determiners describes a reference of a noun.
  - Articles are a subcategory; articles include *the, a*
  - Other determiners include *this, that*
- Adjectives describes properties of nouns.
  - Attributive or adnominal use: *Red* rose, *long* journey, *trendy* magazine
  - Predicative use: The rose is *red*.
  - In some languages, adjectives are inflected to *agree* with nouns.
- Other modifiers of adjectives:
  - *Comparative* (richer, trendier)
  - *Superlative* (richest, trendiest)
  - Some use periphrastic forms (more intelligent, most intelligent) instead of -er and -est suffixes
  - Base case (rich, trendy) is called the *positive* form.

Verbs

Describe actions (*threw* a stone), activities (*walked* along a river), and states (*have* two hands).

Morphological forms:

Base form: walk

Third singular present tense: walks

Gerund and present participle: walking (note gerunds are nouns)

Past tense and past / passive participle: walked

Special verbs: *modal auxiliaries*

Examples: have, should, may, ill

Modals lack some forms of verbs.

Verb Tenses

- The base form (walk) is used in:
  - The present tense: I *walk*
  - The *infinitive*: I like *to walk*.
  - The bare infinitive: She helped me *walk.*
  - Third singular person has different tense form: He *walks*.
- The *-ing* suffix is used in:
  - The *progressive*, which indicates a current action: I am *walking*.
  - The gerund, where verb is treated as a noun: *Walking* is fun.
- The *-ed* form is used in:
  - The past tense: She *walked.*
  - The present perfect: She has *walked.*
  - The past perfect: She had *walked*.

Irregular and Synthetic Verb Forms

Some verbs are *irregular* and have different forms.

She *drove* the car.

She has *driven* a car before.

Some features of verbs can be indicated:

Morphologically or *synthetically*: for example, the suffixes *-s*, *-ed*, *-ing*

Using *auxiliaries* (extra words) to form *verb groups*

Present and past perfect are formed with *have*

Progressive is formed with *be*

Also called periphrastic forms

Adverbs, Prepositions, and Particles

Adverbs modify:

- Verbs: she *often* travels
- Other adverbs or adjectives: *very* fast.
- These latter adverbs are called *degree adverbs* or *qualifiers*.

Prepositions are small words which express spatial relations:

- *In* the glass, *on* the table

Particles are prepositions which can form *phrasal verbs* with verbs:

- Don't *give up.*
- She *ran up* a bill.
- I'm *putting* this *off*.

Conjunctions and Complementizers

Conjunctions combine words.

Coordinating conjunctions relate two words, usually of the same category.

She bought *or* leased.

He *and* I.

Complementizer: a word/morpheme that marks an embedded clause as a complement, typically a subordinating conjunction or infinitival to.

Penn Treebank Tagset

Tag	Description	Example	Tag	Description	Example
CC	coord. conjunction	*and, or*	RB	adverb	*extremely*
CD	cardinal number	*one, two*	RBR	adverb, comparative	*never*
DT	determiner	*a, the*	RBS	adverb, superlative	*fastest*
EX	existential there	*there*	RP	particle	*up, off*
FW	foreign word	*noire*	SYM	symbol	*+, %*
IN	preposition or sub-conjunction	*of, in*	TO	"to"	*to*
JJ	adjective	*small*	UH	interjection	*oops, oh*
JJR	adject., comparative	*smaller*	VB	verb, base form	*fly*
JJS	adject., superlative	*smallest*	VBD	verb, past tense	*flew*
LS	list item marker	*1, one*	VBG	verb, gerund	*flying*
MD	modal	*can, could*	VBN	verb, past participle	*flown*
NN	noun, singular or mass	*dog*	VBP	verb, non-3sg pres	*fly*
NNS	noun, plural	*dogs*	VBZ	verb, 3sg pres	*flies*
NNP	proper noun, sing.	*London*	WDT	wh-determiner	*which, that*
NNPS	proper noun, plural	*Azores*	WP	wh-pronoun	*who, what*
PDT	predeterminer	*both, lot of*	WP$	possessive wh-	*whose*
POS	possessive ending	*'s*	WRB	wh-adverb	*where, how*
PRP	personal pronoun	*he, she*			